OOP Lab Oral Questions - Answers

1. What is Object-Oriented Programming (OOP)?

OOP is a programming approach where programs are organized using objects and classes. It improves reusability, modularity, and security.

2. What are the main principles of OOP?

Encapsulation, Inheritance, Polymorphism, and Abstraction.

3. What is a class and an object in Java?

A class is a blueprint, while an object is an instance of a class.

4. Difference between constructor and method?

Constructor initializes objects and has same name as class. Methods perform tasks and can have any valid name.

5. What is constructor overloading?

Using multiple constructors with different parameters in the same class.

6. What is method overloading?

Using methods with same name but different parameters.

7. Role of this keyword in Java?

It refers to the current object.

8. Difference between static and non-static members?

Static belongs to class; non-static belongs to object.

9. What are access modifiers in Java?

public, private, protected, and default.

10. What is encapsulation?

Wrapping data and methods together and hiding data using private variables.

11. What is inheritance?

Inheritance allows one class to acquire properties of another class.

12. What is single inheritance?

One child class inherits one parent class.

13. What is method overriding?

Child class provides its own implementation of parent class method.

14. What is runtime polymorphism?

Polymorphism achieved using method overriding at runtime.

15. What is compile-time polymorphism?

Polymorphism achieved using method overloading.

16. What is an abstract class?

A class declared with abstract keyword that cannot be instantiated.

17. What is an interface in Java?

An interface contains abstract methods and constants.

18. Difference between abstract class and interface?

Abstract class can contain normal methods; interface mainly contains abstract methods.

19. Can a class implement multiple interfaces?

Yes, Java supports multiple inheritance through interfaces.

20. What is dynamic method dispatch?

Method call is resolved at runtime.

21. What is exception handling in Java?

Mechanism to handle runtime errors without crashing the program.

22. Difference between checked and unchecked exceptions?

Checked exceptions are checked at compile time; unchecked at runtime.

23. Purpose of try-catch-finally block?

try contains risky code, catch handles exceptions, finally executes always.

24. What happens if exception is not handled?

Program terminates abnormally.

25. Use of throw and throws keywords?

throw manually throws exception; throws declares exceptions.

26. Difference between throw and throws?

throw is used inside method; throws is used in method declaration.

27. How to handle multiple exceptions?

Using multiple catch blocks or multi-catch.

28. Why is finally block important?

It executes whether exception occurs or not.

29. What is a thread in Java?

A lightweight subprocess used for parallel execution.

30. Difference between process and thread?

Process is heavyweight; thread is lightweight.

31. What is thread lifecycle?

New, Runnable, Running, Waiting/Blocked, Terminated.

32. What is synchronization in Java?

Controls access of multiple threads to shared resources.